WORKED SOLUTION

STEP II 1988, Question 4

Pure • Loci in the Argand Diagram

STEP II 1988, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q4

The key idea

Write the complex number as $a + ib$ and turn "real" or "imaginary" into one real equation in a and b. That gives a familiar curve in the Argand diagram. Then squaring is a map of the plane, and we find where it sends that curve by writing x and y (or p and q) in terms of a and b and eliminating.

A word on the printed text: it says $w^2 - 2x$, but x has not been introduced at that point, and the final part (with $t^2 - 2t$) shows the intended condition. We read it as $w^2 - 2w$ *is real*.

Solution

Part (i)

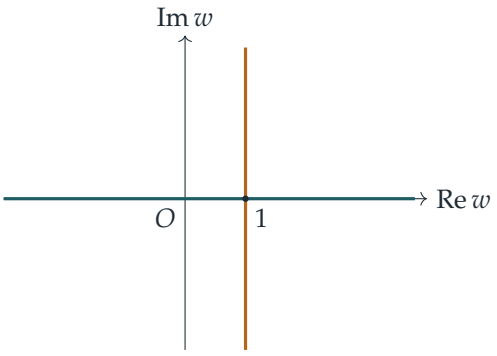
Let $w = a + \mathrm{i}b$ with a, b real. Then

$$w^2 - 2w = (a^2 - b^2 - 2a) + \mathrm{i}\,(2ab - 2b) = (a^2 - b^2 - 2a) + 2\mathrm{i}\,b(a - 1).$$

This is real exactly when its imaginary part vanishes, so when

$$b(a - 1) = 0, \qquad \text{that is,} \qquad b = 0 \ \text{ or } \ a = 1.$$

The locus of w is therefore the real axis together with the vertical line $\operatorname{Re} w = 1$. The two lines cross at $w = 1$.



Part (ii)

We square each of the two lines separately.

The real axis. If $b = 0$ then $w = a$ is real and $w^2 = a^2$. So $y = 0$ and $x = a^2 \geq 0$, and as a runs through the reals, a^2 takes every value in $[0, \infty)$. This piece of the locus is the non-negative x-axis: the half-line $y = 0$, $x \geq 0$.

The line $a = 1$. If $w = 1 + ib$ then

$$w^2 = 1 - b^2 + 2ib, \qquad \text{so} \qquad x = 1 - b^2, \quad y = 2b.$$

Eliminating $b = \frac{1}{2}y$ gives

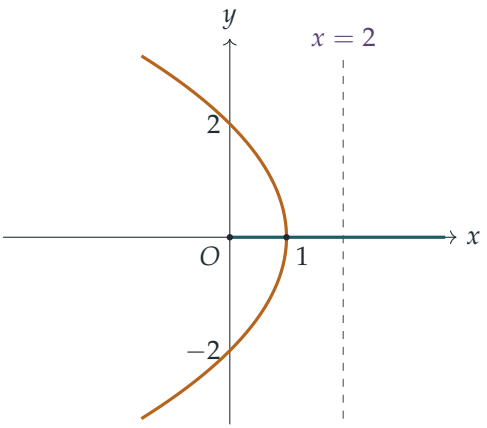
$$x = 1 - \tfrac{1}{4}y^2, \qquad \text{or} \qquad y^2 = 4(1 - x) = -4(x - 1).$$

Every point of this curve is reached, by taking $b = \frac{1}{2}y$. It is a parabola with its axis along the x-axis, vertex at $(1, 0)$, opening towards negative x. Comparing with the standard form $Y^2 = 4cX$, here $c = -1$ with $X = x - 1$. So the focus is at $X = -1$, which is the origin, and the directrix is $X = 1$, which is the line $x = 2$. Indeed $x^2 + y^2 = x^2 + 4 - 4x = (2 - x)^2$, so every point is as far from O as from the line $x = 2$. The parabola cuts the y-axis at $(0, \pm 2)$.

Ask yourself Why doesn't the half-line extend to negative x?

The real axis in the w-plane is covered twice (by a and $-a$), but squaring a real number never gives a negative result. So the whole real axis folds onto the non-negative half. Similarly, the line $\mathrm{Re}\, w = 1$ goes onto the parabola once, because $b \mapsto 2b$ is one-to-one.

So the locus of (x, y) is the parabola $y^2 = 4(1 - x)$, with focus at the origin and directrix $x = 2$, together with the half-line $y = 0$, $x \geq 0$. The half-line runs from the focus through the vertex $(1, 0)$, which is the image of $w = 1$, where the two lines of part (i) meet.



The final part

Let $t = a + \mathrm{i}b$. "Imaginary" here means purely imaginary, so we need the real part of $t^2 - 2t$ to vanish:

$$a^2 - b^2 - 2a = 0, \qquad \text{that is,} \qquad (a-1)^2 - b^2 = 1.$$

So t lies on a rectangular hyperbola centred at 1, with vertices at $t = 0$ and $t = 2$. On it, $b^2 = a^2 - 2a = a(a-2) \geq 0$, so $a \leq 0$ or $a \geq 2$.

Now $t^2 = p + \mathrm{i}q$ gives

$$p = a^2 - b^2 = 2a, \qquad q = 2ab,$$

where we used the hyperbola's equation $a^2 - b^2 = 2a$ to simplify p. So $a = \frac{1}{2}p$, and then

$$q^2 = 4a^2b^2 = 4a^2(a^2 - 2a) = p^2\left(\tfrac{1}{4}p^2 - p\right), \qquad \text{so} \qquad 4q^2 = p^3(p-4).$$

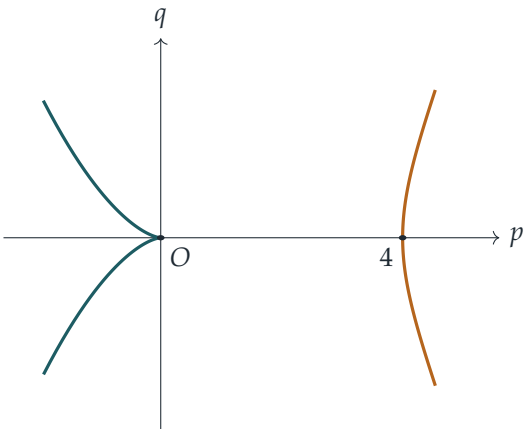
The conditions $a \leq 0$ or $a \geq 2$ become $p \leq 0$ or $p \geq 4$, which are exactly where $p^3(p-4) \geq 0$. Conversely, take a point on this curve. If $p \neq 0$, put $a = \frac{1}{2}p$ and $b = q/p$. Then $b^2 = q^2/p^2 = \frac{1}{4}p(p-4) = a^2 - 2a$, so t is on the hyperbola, and $t^2 = (a^2 - b^2) + 2\mathrm{i}ab = p + \mathrm{i}q$. If $p = 0$ then $q = 0$, and this point comes from $t = 0$. So the locus is the whole curve

$$4q^2 = p^3(p-4).$$

Its shape:

- It is symmetric in the p-axis, because only q^2 appears.
- There are no points with $0 < p < 4$. The curve has two pieces, one coming from each branch of the hyperbola.
- The right piece starts at $(4, 0)$, the image of $t = 2$. Near there $q^2 \approx 16(p-4)$, so the tangent is vertical.
- The left piece passes through the origin, the image of $t = 0$. Near there $q^2 \approx -p^3$, so $|q| \approx |p|^{3/2}$, a cusp pointing right, with both arms tangent to the p-axis.

- For large $|p|$, $|q| \approx \frac{1}{2}p^2$, so both pieces open out like parabolas.



(The vertical scale is compressed. The teal piece comes from the branch of the hyperbola with $\operatorname{Re} t \leq 0$, the amber piece from the branch with $\operatorname{Re} t \geq 2$.)

Checks

- Part (i): $w = 1 + 2\mathrm{i}$ gives $w^2 - 2w = (-3 + 4\mathrm{i}) - (2 + 4\mathrm{i}) = -5$, which is real. Part (ii): its square $-3 + 4\mathrm{i}$ satisfies $16 = 4(1 - (-3))$.
- Final part: $t = -1 + \mathrm{i}\sqrt{3}$ satisfies $1 - 3 + 2 = 0$. Its square is $-2 - 2\sqrt{3}\,\mathrm{i}$, and $4q^2 = 48 = (-8)(-6) = p^3(p - 4)$.
- We checked numerically at thousands of random points that each locus maps onto the stated curve, and that every point of each curve is reached.

Common slips

- Forgetting the real axis in part (i), and so missing the half-line in part (ii). Or drawing the whole x-axis instead of only $x \geq 0$.
- Not using $a^2 - b^2 = 2a$ to get $p = 2a$. Without it, the elimination is much messier.
- Missing the gap $0 < p < 4$, or the cusp at the origin, in the final sketch.

Beyond STEP

Squaring, $z \mapsto z^2$, is the simplest conformal map. It doubles angles, which is why the lines and hyperbolas here turn into parabolas and cusped curves. At $z = 0$ the derivative vanishes, so angles are no longer preserved, and that is where the cusp comes from. A standard trick in fluid flow and electrostatics is to use maps like this to

turn a hard region into an easy one. Needham's *Visual Complex Analysis* explains these mappings beautifully with pictures.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.